

M6 — Raycast + sleeping / islands

Status: planned.

Goal

Cover the remaining engine subsystems the game touches: downward raycast, the sleep/island machinery, and the kinematic-mover wake hack.

Scope

- **Raycast** — straight-down `Ray` + `RayResult` (`raycastFloorY`), with the interaction filter.
- **Sleeping / islands** — body sleep thresholds, island construction, and the order islands/bodies are processed (affects solver determinism).
- `wakeJointPartners` — the zero-impulse wake trick for sleeping welded riders of a kinematic mover (`NapeWorld.hx:369`).
- `touchingDynamicBodies` (arbiter scan), `bodyContains` , `bodyArea` .

Gate

- Raycast hit distance/point matches the reference bit-for-bit across configs.
- A scene that settles and **sleeps** matches the reference step-for-step, including *when* each body sleeps and the post-sleep stillness.
- The **coin-sensor** and switch-persist scenarios stay bit-identical.

Risks / open questions

- Sleep timing is threshold-sensitive; a one-frame-early sleep diverges the whole scene. The multi-step gate localizes it.